

Executive Summary – Telecom Customer Churn Analysis

Business Objective

The objective of this analysis was to identify the factors associated with customer churn and uncover actionable insights that can help the telecom company improve customer retention.

The dataset contains **7,043 customers**, with information on demographics, services subscribed, contract details, billing preferences, payment methods, tenure, and churn status.

Dataset Quality Assessment

Before analysis, the dataset was thoroughly cleaned and validated.

Data Preparation Steps

- Converted **TotalCharges** to numeric format.
- Replaced blank values with 0 where necessary.
- Checked for missing values and found none.
- Verified that there were no duplicate records.
- Examined distributions and summary statistics of numerical variables.

Key Numerical Statistics

Variable	Mean	Median	Min	Max
Tenure	32.37 months	29 months	0	72
Monthly Charges	\$64.76	\$70.35	\$18.25	\$118.75
Total Charges	\$2,279.73	\$1,394.55	\$0	\$8,684.80

The average customer remains with the company for approximately **32 months**, indicating a relatively mature customer base.

Overall Churn Analysis

Customer Retention vs Churn

- **73.46%** of customers remain with the company.
- **26.54%** of customers have churned.

This means that approximately **1 out of every 4 customers leaves the service**, representing a substantial loss of recurring revenue and customer lifetime value.

Key Insight

While the majority of customers remain loyal, a churn rate of more than 25% suggests significant opportunities for improving customer retention strategies.

Customer Demographics

Gender and Churn

The churn distribution between male and female customers is relatively similar.

Insight

- Both genders exhibit comparable churn behavior.
- Gender does not appear to be a strong predictor of customer attrition.

Business Implication

Retention strategies should focus on customer behavior, service usage, and contract characteristics rather than gender-based segmentation.

Senior Citizens and Churn

Analysis shows that the proportion of churned customers is noticeably higher among senior citizens than among non-senior customers.

Insight

Senior citizens represent a more vulnerable customer segment and appear more likely to discontinue services.

Possible Reasons

- Difficulty adapting to technology-related services.
- Higher price sensitivity.
- Greater need for customer support.

Business Recommendation

Develop targeted retention programs for senior customers through:

- Dedicated support channels.
 - Simplified service plans.
 - Loyalty discounts.
 - Personalized onboarding assistance.
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Tenure Analysis

Customer Lifetime and Churn

The tenure distribution reveals a strong relationship between customer longevity and churn.

Findings

- Customers who churn generally have very low tenure.
- Churn is concentrated within the first few months of subscription.
- Long-tenure customers are significantly more likely to remain with the company.

Insight

The highest-risk period occurs during the early stages of the customer lifecycle.

Business Recommendation

Implement a structured onboarding strategy during the first 3–6 months:

- Welcome campaigns.
- Service education.
- Early customer satisfaction surveys.
- Proactive customer support.

Reducing early-stage churn can significantly improve overall retention.

Contract Type Analysis

Contract Duration and Churn

Contract type emerges as one of the strongest predictors of customer retention.

Findings

Customers with:

- **Month-to-month contracts** show the highest churn.
- **One-year contracts** exhibit substantially lower churn.
- **Two-year contracts** demonstrate the strongest retention.

Insight

Long-term contractual commitments significantly reduce customer attrition.

Business Interpretation

Customers on month-to-month contracts can leave with little switching cost, whereas longer contracts encourage customer loyalty.

Business Recommendation

Encourage migration from month-to-month plans by offering:

- Contract renewal incentives.
 - Bundled discounts.
 - Loyalty rewards.
 - Reduced pricing for annual commitments.
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Service Subscription Analysis

The analysis examined the relationship between churn and telecom service adoption.

Services Associated with Lower Churn

Customers who remain with the company are more likely to subscribe to:

- Phone Service
- Internet Service
- Online Security

- Device Protection
- Tech Support

Insight

Customers using multiple services become more integrated into the company's ecosystem and therefore have lower propensity to leave.

Online Security

Customers without Online Security exhibit noticeably higher churn rates.

Interpretation

Security-related services increase perceived value and dependence on the provider.

Online Backup

Customers who do not use Online Backup services show higher churn tendencies.

Interpretation

Additional utility services create stronger customer attachment.

Device Protection

Customers subscribed to Device Protection plans are generally more likely to remain with the company.

Interpretation

Protection services increase customer confidence and satisfaction.

Tech Support

One of the clearest churn indicators.

Findings

Customers without Tech Support churn considerably more frequently.

Interpretation

Access to support improves customer experience and reduces frustration.

Streaming Services

Customers who subscribe to Streaming TV and Streaming Movies tend to remain longer than customers who do not utilize these services.

Interpretation

Entertainment services increase customer engagement and overall value perception.

Payment Method Analysis

Payment Method and Churn

Payment method exhibits a strong association with customer attrition.

Key Finding

Customers using **Electronic Check** have the highest churn tendency among all payment methods.

Possible Reasons

- Greater payment friction.
- Less automated billing experience.
- Lower customer commitment.

Customers using:

- Automatic bank transfers
- Automatic credit card payments

show better retention performance.

Business Recommendation

Encourage customers to switch to automated payment methods through:

- Cashback offers.
 - Billing discounts.
 - Loyalty rewards.
 - Easier payment setup.
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Strategic Business Recommendations

Based on the findings, the company should prioritize the following initiatives:

1. Reduce Early-Tenure Churn

Focus retention efforts on customers within their first six months.

2. Promote Long-Term Contracts

Offer incentives to move customers away from month-to-month contracts.

3. Increase Adoption of Value-Added Services

Promote:

- Online Security
- Online Backup
- Device Protection
- Tech Support

These services are associated with stronger retention.

4. Improve Senior Citizen Retention

Develop customer support programs specifically tailored for older customers.

5. Encourage Automatic Payments

Reduce reliance on Electronic Check payments.

6. Bundle Services

Create bundled offerings that combine internet, security, support, and entertainment services.

Conclusion

This analysis demonstrates that customer churn is primarily driven by **tenure, contract type, service adoption, senior citizen status, and payment method** rather than demographic characteristics such as gender.

The most important risk factors identified are:

- Being a new customer
- Having a month-to-month contract
- Not subscribing to support/security services
- Using Electronic Check payments
- Belonging to the senior citizen segment

With an overall churn rate of **26.54%**, targeted retention strategies focused on these high-risk groups can significantly improve customer loyalty, customer lifetime value, and long-term revenue growth.